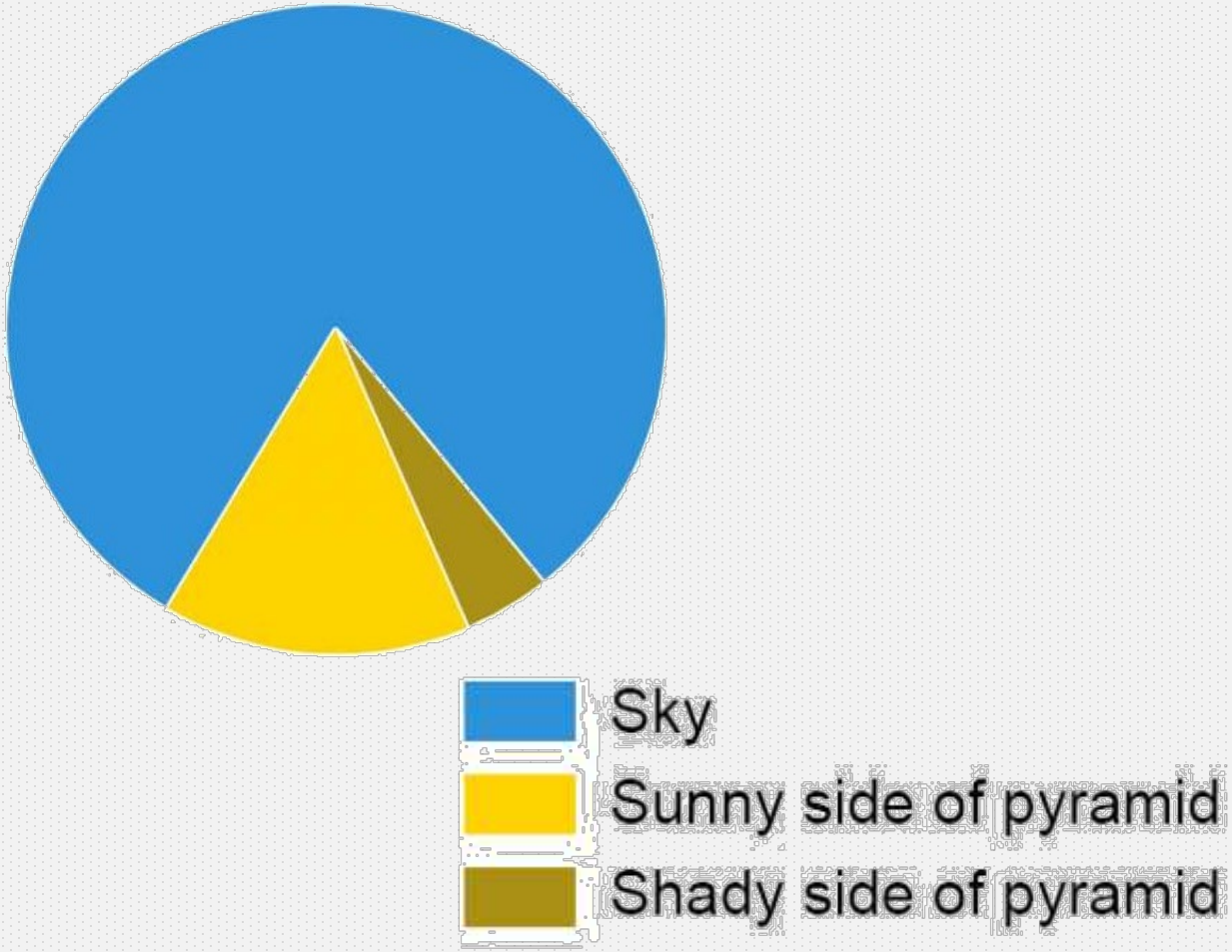
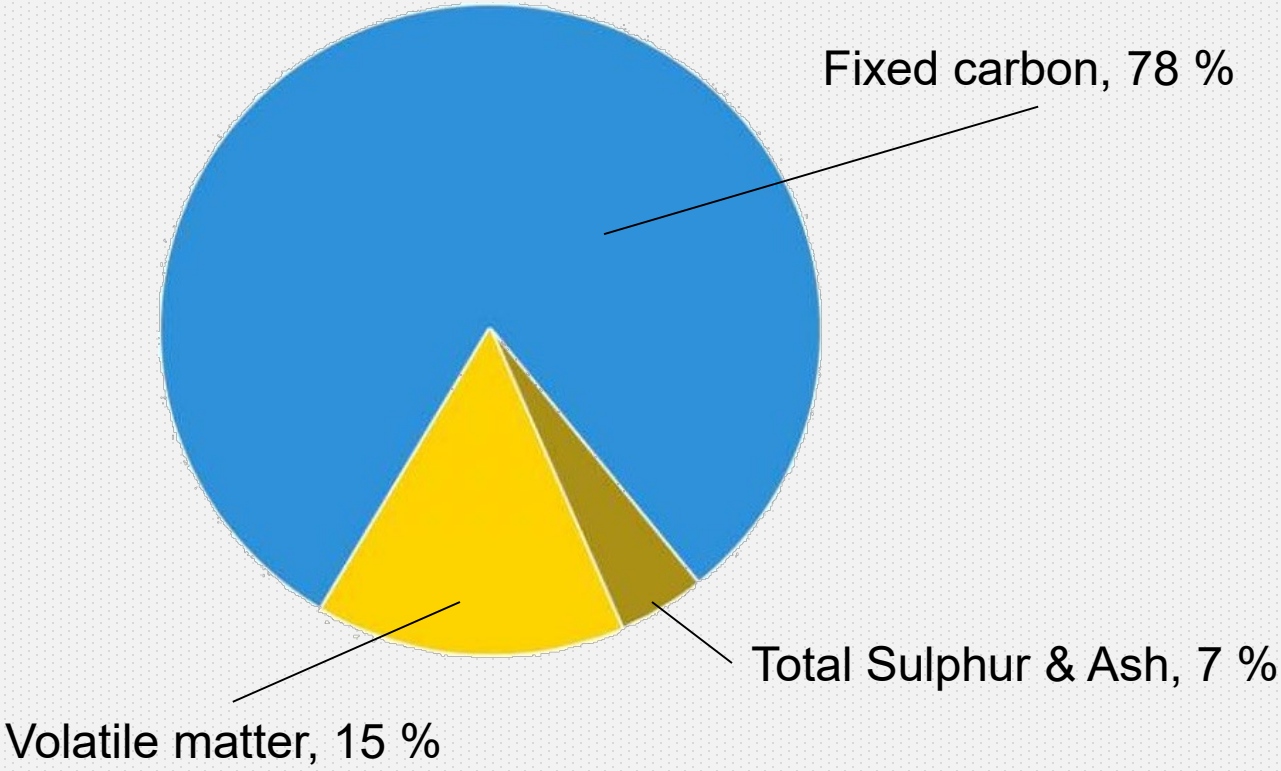
OCTOBER	TOPIC
23	**DESCRIPTIVE STATISTICS (1/2)** **Displaying data** Stem and Leaf Diagrams Frequency Histograms Relative Frequency Histograms **Measures of Central Location** The Mean The Median The Mode **Measures of Variability** The Range The Variance and the Standard Deviation


Sky
Sunny side of pyramid
Shady side of pyramid

Proximate analysis of bituminous coal sample:



DISPLAYING DATA

- Suppose 30 students in a statistics class took a test and made the following scores:

86, 80, 25, 77, 73, 76, 100, 90, 69, 93, 90, 83, 70, 73, 73, 70, 90, 83, 71, 95, 40, 58, 68, 69, 100, 78, 87, 97, 92, 74

How did the class do on the test?

Stem and Leaf Diagrams

86, 80, 25, 77, 73, 76, 100, 90, 69, 93, 90, 83, 70, 73, 73, 70, 90, 83, 71, 95, 40, 58, 68, 69, 100, 78, 87, 97, 92, 74

The general purpose of a stem and leaf diagram is to provide a quick display of how the data are distributed across the range of their values.

Stem and Leaf Diagrams

86, 80, 25, 77, 73, 76, 100, 90, 69, 93, 90, 83, 70, 73, 73, 70, 90, 83, 71, 95, 40, 58, 68, 69, 100, 78, 87, 97, 92, 74

The numbers in the tens place, from 2 through 9, and additionally the number 10, are the "**stems**," and are arranged in numerical order from top to bottom to the left of a vertical line. The number in the units place in each measurement is a "**leaf**," and is placed in a row to the right of the corresponding stem, the number in the tens place of that measurement

Stem and Leaf Diagrams

86, 80, 25, 77, 73, 76, 100, 90, 69, 93, 90, 83, 70, 73, 73, 70, 90, 83, 71, 95, 40, 58, 68, 69, 100, 78, 87, 97, 92, 74

1										
2	5									
3										
4	0									
5	8									
6	8	9	9							
7	0	0	1	3	3	3	4	6	7	8
8	0	3	3	6	7					
9	0	0	2	3	5	7				
10	0	0								

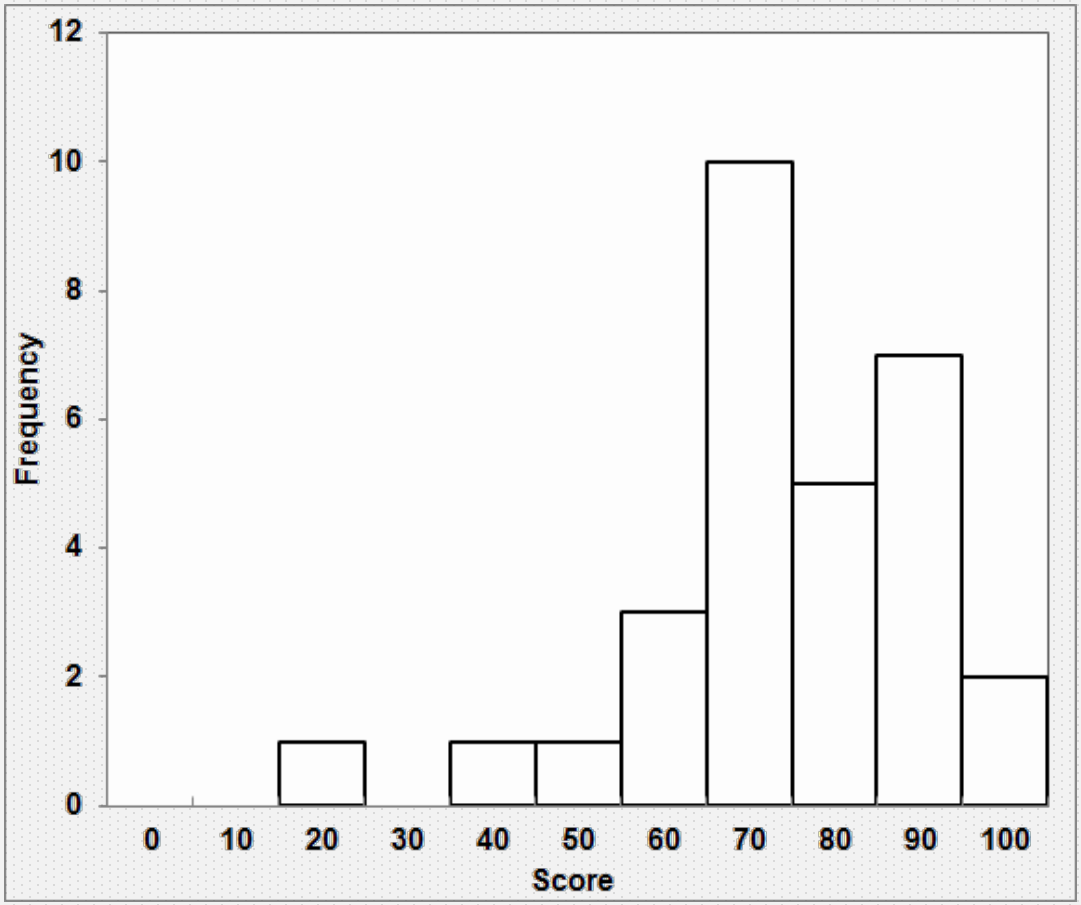
How to represent large data?

		X_co-ordinate	Y_co-ordinate	Z_co-ordinate	Cu (%)	Mo (%)	core_length	from	to
1	E2S	4301	5192.97	5179.44	0.615	0.05	5	10	15
2	E2S	4301	5188.43	5170.53	0.513	0.101	15	15	30
3	E2S	4301	5181.62	5157.17	0.558	0.057	15	30	45
4	E2S	4301	5174.81	5143.80	0.658	0.089	15	45	60
5	E2S	4301	5168.00	5130.44	0.57	0.037	15	60	75
6	E2S	4301	5161.19	5117.07	0.523	0.037	15	75	90
7	E2S	4301	5154.38	5103.71	0.55	0.035	15	90	105
8	E2S	4301	5147.57	5090.34	0.592	0.034	15	105	120
.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.
1794	K65S19	1916.4	3181.34	4771.14	0.628	0.055	15	495	510
1795	K65S19	1916.4	3174.53	4757.77	0.667	0.046	15	510	525
1796	K65S19	1916.4	3167.72	4744.41	0.628	0.067	15	525	540
1797	K65S19	1916.4	3160.91	4731.04	0.569	0.034	15	540	555
1798	K65S19	1916.4	3154.10	4717.68	0.611	0.043	15	555	570
1799	K65S19	1916.4	3147.29	4704.31	0.704	0.046	15	570	585
1800	K65S19	1916.4	3140.48	4690.95	0.795	0.112	15	585	600

Source: http://www.kriging.com/datasets/PG2000_big_data_files.zip

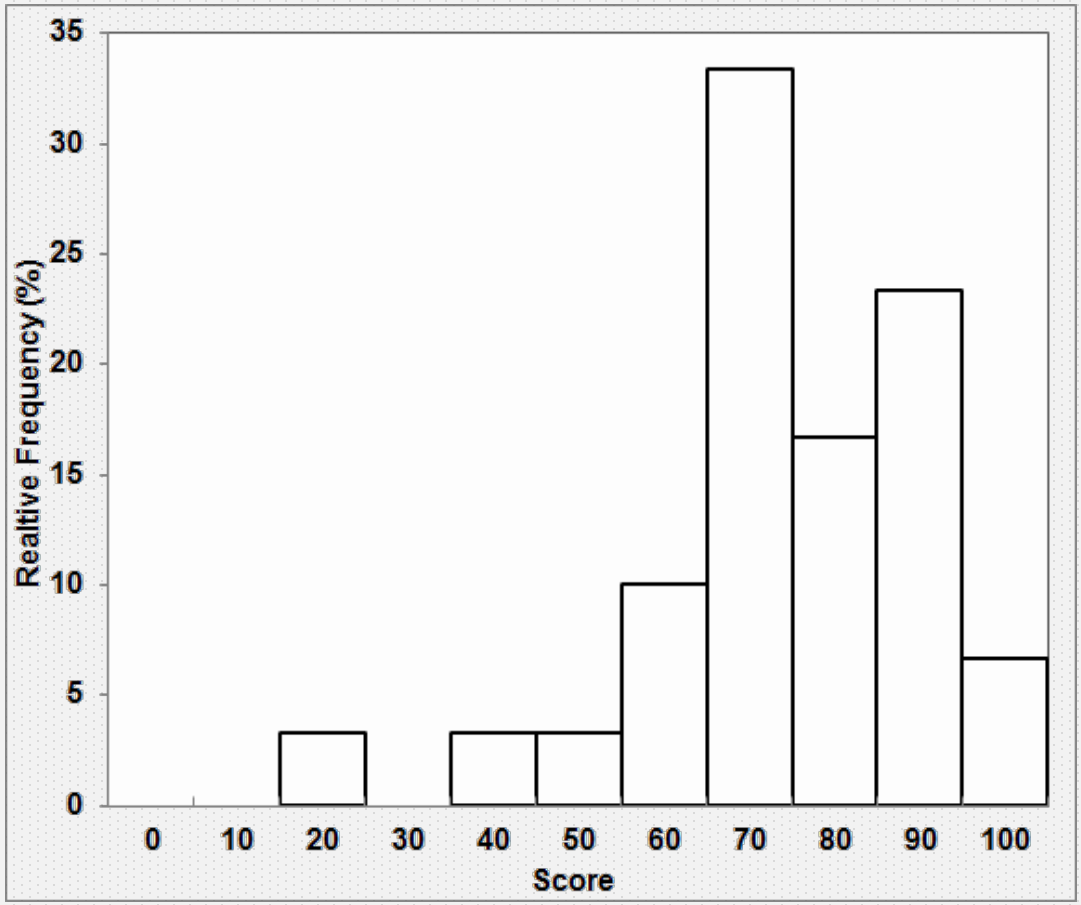
Frequency Histograms

86, 80, 25, 77, 73, 76, 100, 90, 69, 93, 90, 83, 70, 73, 73, 70, 90, 83, 71, 95, 40, 58, 68, 69, 100, 78, 87, 97, 92, 74

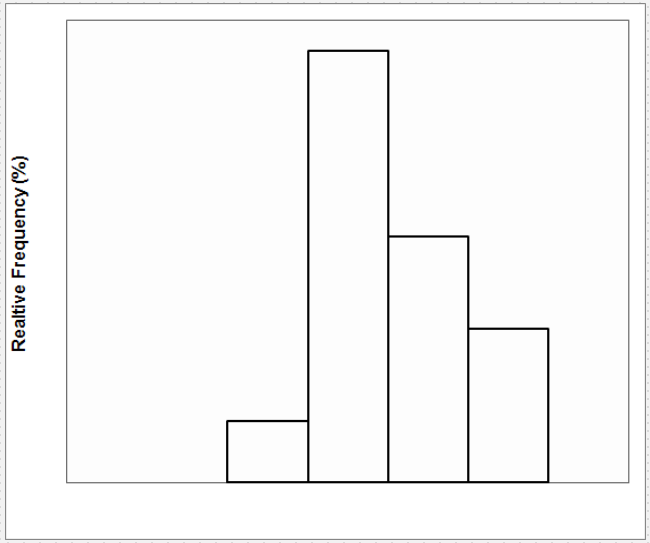


Relative Frequency Histograms

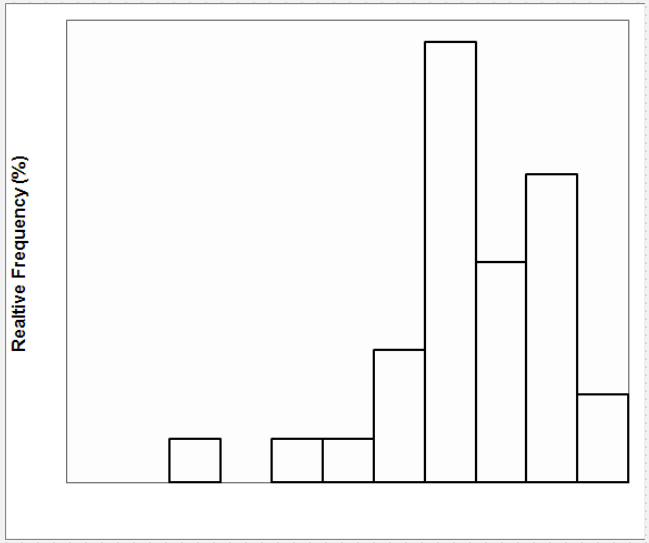
86, 80, 25, 77, 73, 76, 100, 90, 69, 93, 90, 83, 70, 73, 73, 70, 90, 83, 71, 95, 40, 58, 68, 69, 100, 78, 87, 97, 92, 74



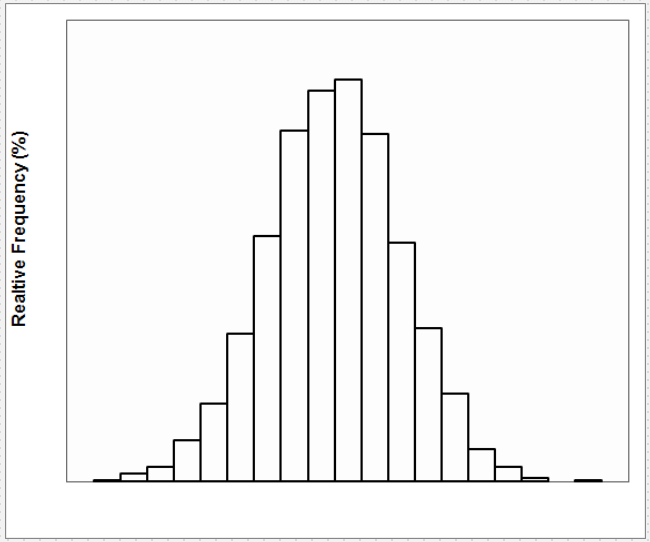
Small Sample



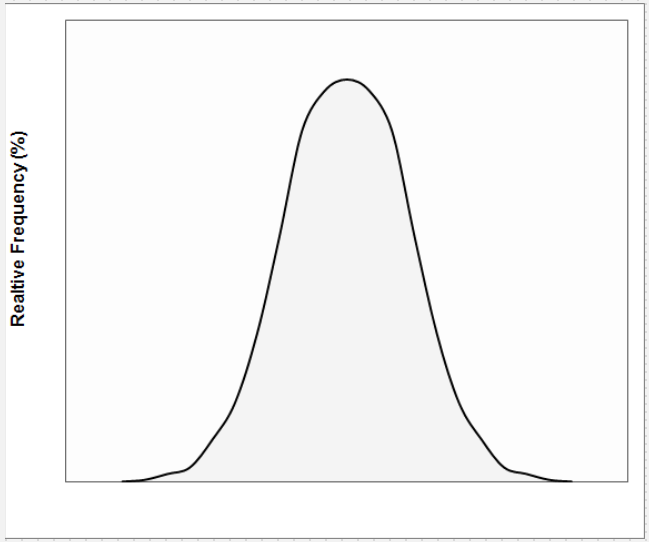
Medium Sample

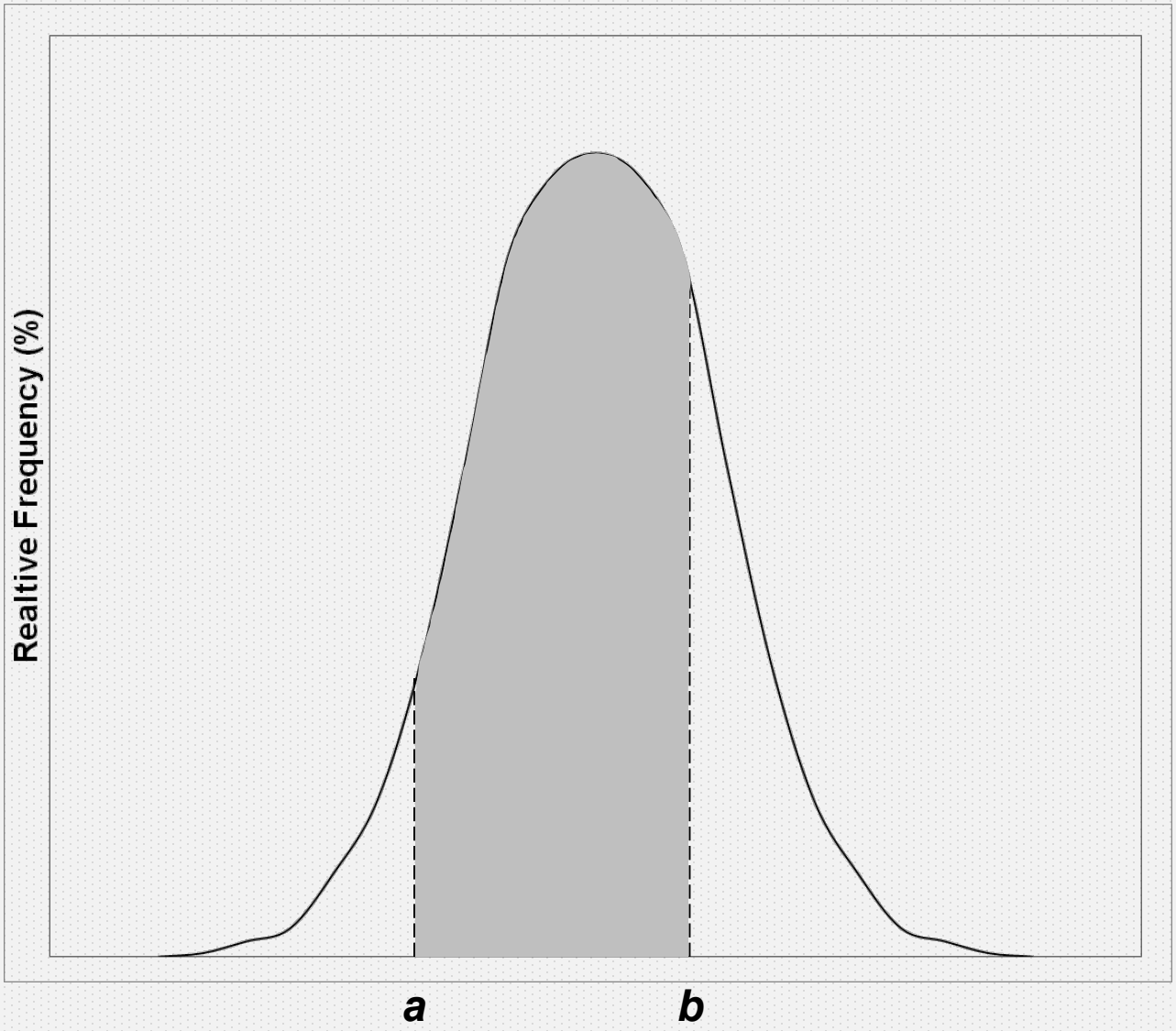


Large Sample



Very Very Large Sample





SHADED AREA = Proportion of Data between *a* and *b*

KEY TAKEAWAYS

- For a very large sample the relative frequency histogram would look very fine. If the sample size were to increase indefinitely then the corresponding relative frequency histogram would be so fine that it would look like a smooth curve.

(It is good to keep in mind that such a curve is actually just a very fine relative frequency histogram in which the exceedingly narrow vertical bars have disappeared.)

- Because the area of each such vertical bar is the proportion of the data that lies in the interval of numbers over which that bar stands, this means that for any two numbers a and b, the proportion of the data that lies between the two numbers a and b is the area under the curve that is above the interval (a,b) in the horizontal axis

MEASURES OF CENTRAL LOCATION

- Any kind of “average” is meant to be an answer to the question “Where do the data center?” It is thus a measure of the central location of the data set.

- Different shapes of the histogram call for different measures of central location.

Recall

86, 80, 25, 77, 73, 76, 100, 90, 69, 93, 90, 83, 70, 73, 73, 70, 90, 83, 71, 95, 40, 58, 68, 69, 100, 78, 87, 97, 92, 74

Find $\sum x$, $\sum x^2$, and $\sum (x-1)^2$

$$\sum x = 86 + 80 + 25 + \cdots + 97 + 92 + 74 = 2330$$

$$\sum x^2 = 86^2 + 80^2 + 25^2 + \cdots + 97^2 + 92^2 + 74^2 = 188818$$

$$\sum (x-1)^2 = 85^2 + 79^2 + 24^2 + \cdots + 96^2 + 91^2 + 73^2 = 184188$$

Sample Mean

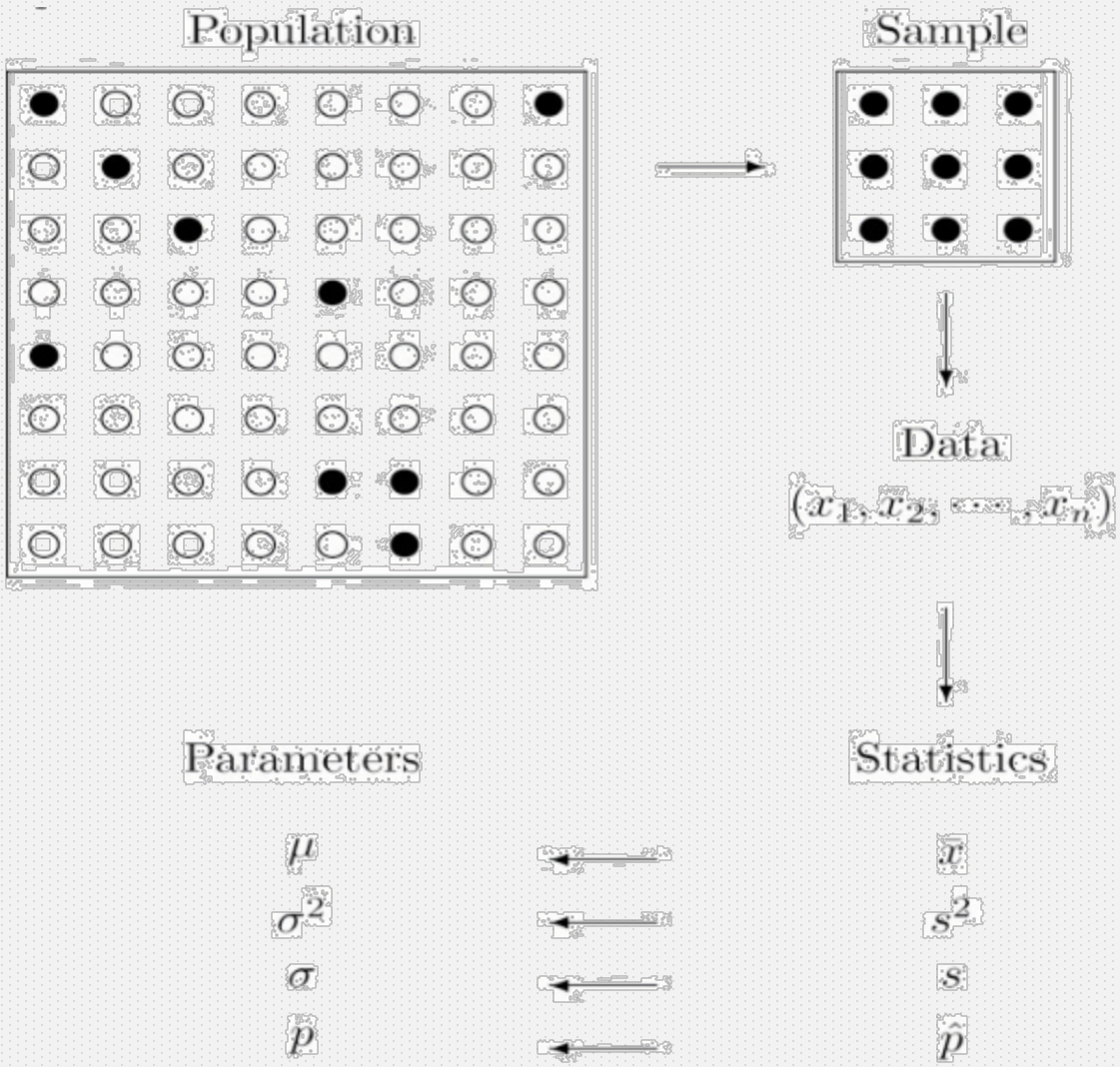
For a set of n sample data, it is denoted by $\overline{x}$ and defined by:

$$\overline{x} = \frac{\sum x}{n}$$

Population Mean

For a set of N population data, it is denoted by μ and defined by:

$$\mu = \frac{\sum x}{N}$$


Population
Sample
Data
$(x_1, x_2, \ldots, x_n)$
Parameters
Statistics
μ
$\bar{x}$
σ^2
s^2
σ
s
p
$\hat{p}$

Random sample of seven employees

Average yearly income of employees at a large corporation:

($/yr) (× 1000)
24.8
22.8
24.6
192.5
25.2
18.5
23.7

Sample Mean

24.8, 22.8, 24.6, 192.5, 25.2, 18.5, 23.7

$$\bar{x} = \frac{\sum x}{n} = \frac{24.8 + 22.8 + 24.6 + 192.5 + 25.2 + 18.5 + 23.7}{7}$$

$$\bar{x} = 47.4$$

"the average income of employees at this corporation is $47,400"

surely misleading

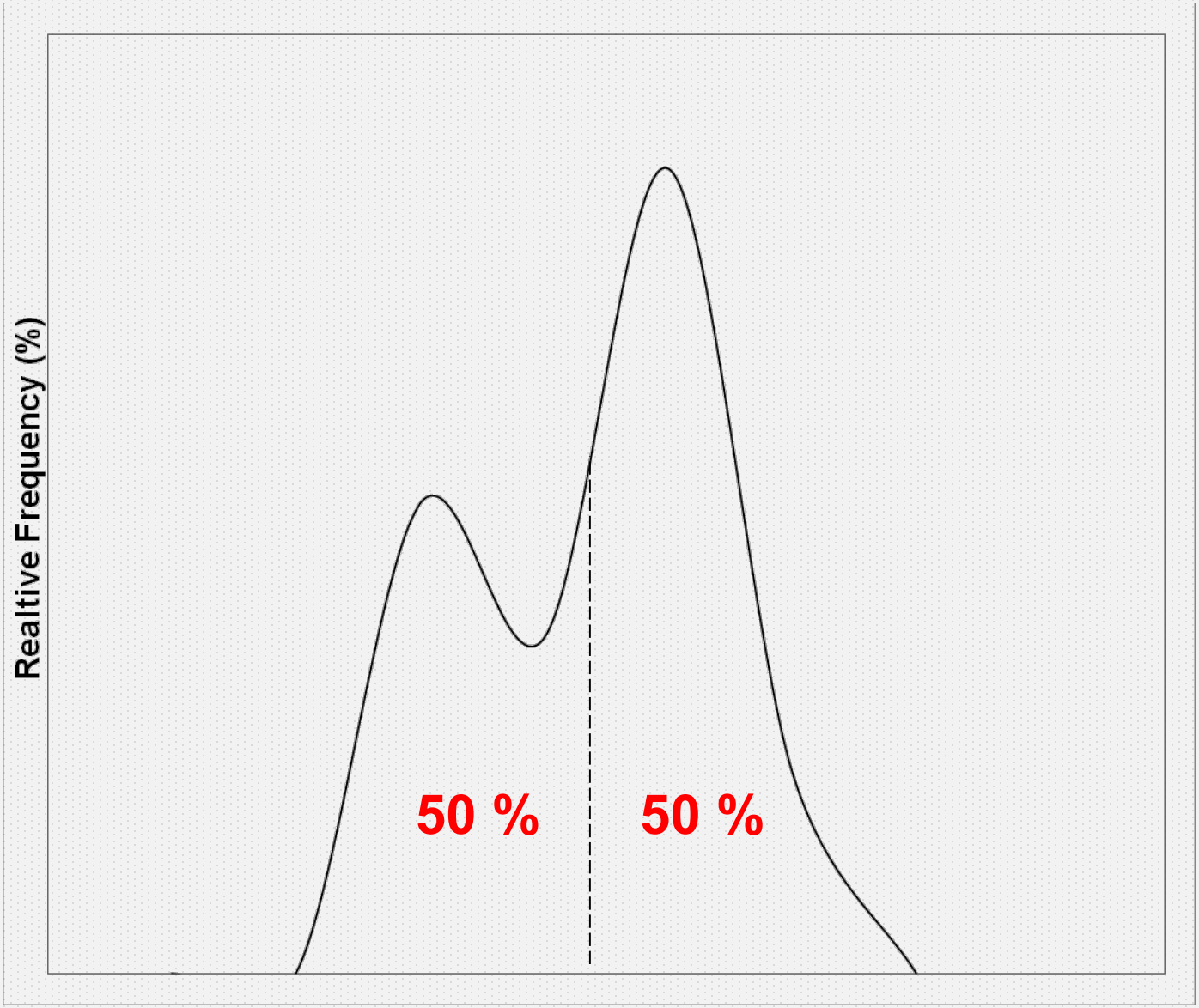
The Median

A better measure of the “center” of the data if we were to arrange the data in numerical order,

18.5 22.8 23.7 24.6 24.8 25.2 192.5

then select the middle number in the list, in this case **24.6**

The Median



The Median

Find the median

3.00, 3.71, 1.90, 1.39, 1.76, 4.00, 2.53, 2.71

First step is to sort. So,

1.39, 1.76, 1.90, 2.53, 2.71, 3.00, 3.71, 4.00

Median

$$\frac{2.53 + 2.71}{2} = 2.62$$

Number of courses students added per semester

6, 5, 7, 5, 4, 7, 6, 4, 5

$$\bar{x} = \frac{\sum x}{n} = \frac{6+5+7+5+4+7+6+4+5}{9}$$

$$\bar{x} = 5.44$$

"the average number of courses student added at semester is 5.44"

surely non-sense

The Mode

The sample mode of a set of sample data is the most frequently occurring value.

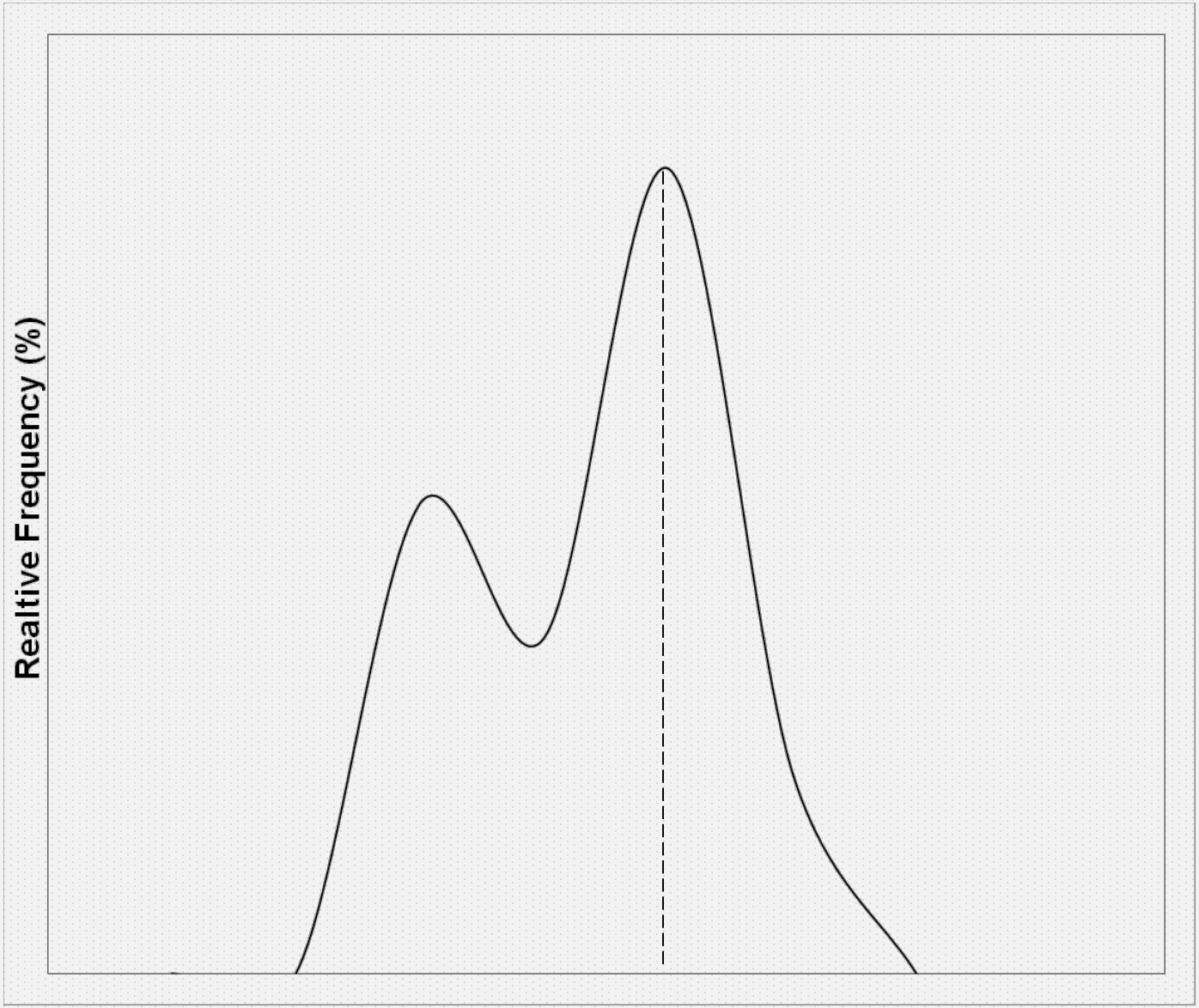
6, 5, 7, 5, 4, 7, 6, 4, 5

Unique values are **{4, 5, 6, 7}**

Occuring Frequencies are **{2, 3, 2, 2}**

So, the mode of data set is “5”

The Mode



Mode

KEY TAKEAWAYS

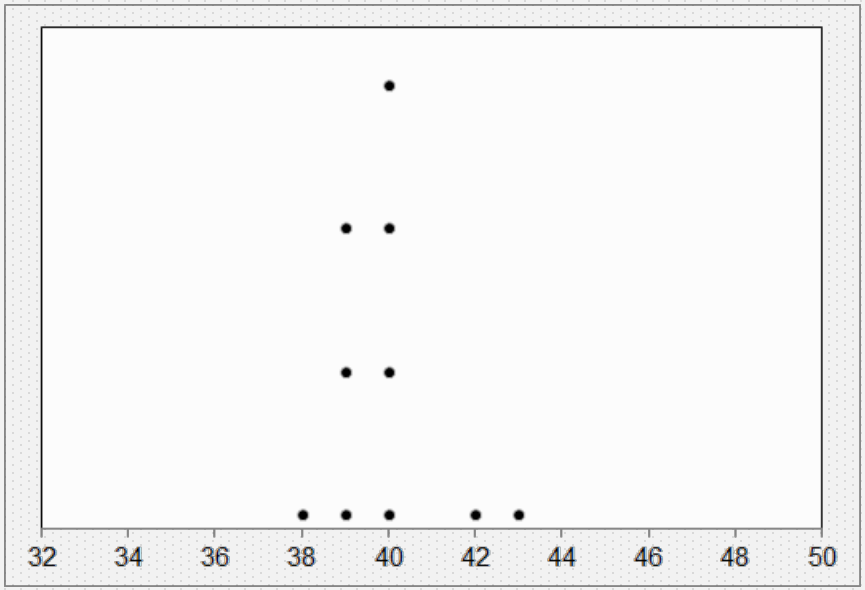
- The mean, the median, and the mode each answer the question "Where is the center of the data set?"

- The nature of the data set, as indicated by a relative frequency histogram, determines which one gives the best answer.

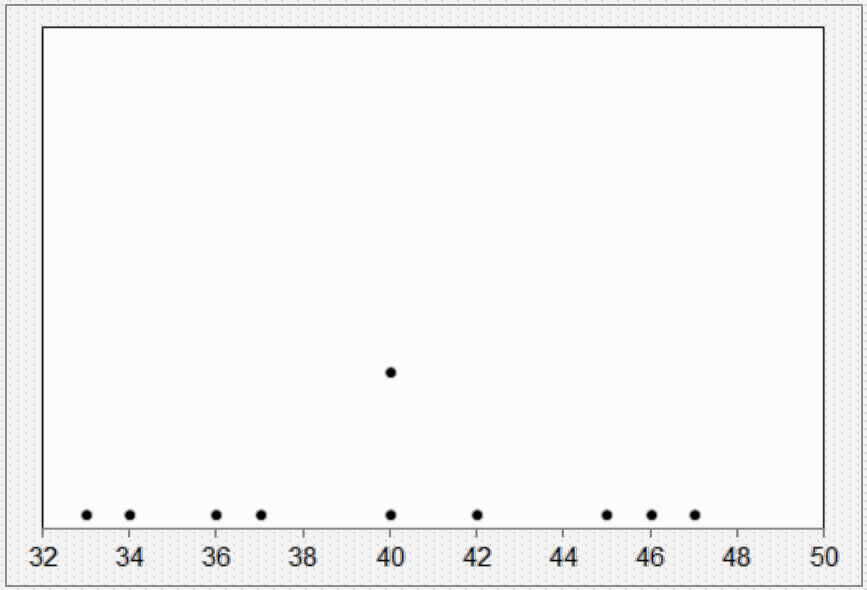
MEASURES OF VARIABILITY

Data Set 1:	**40**	**38**	**42**	**40**	**39**	**39**	**43**	**40**	**39**	**40**
Data Set 2:	**46**	**37**	**40**	**33**	**42**	**36**	**40**	**47**	**34**	**45**

They both have mean, median, and mode (40). How to characterize them?



Data Set 1

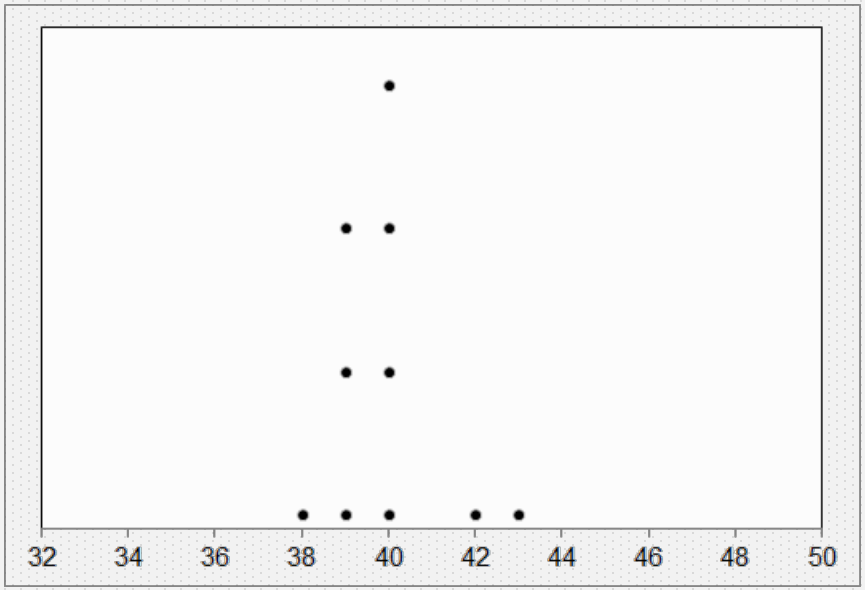


Data Set 2

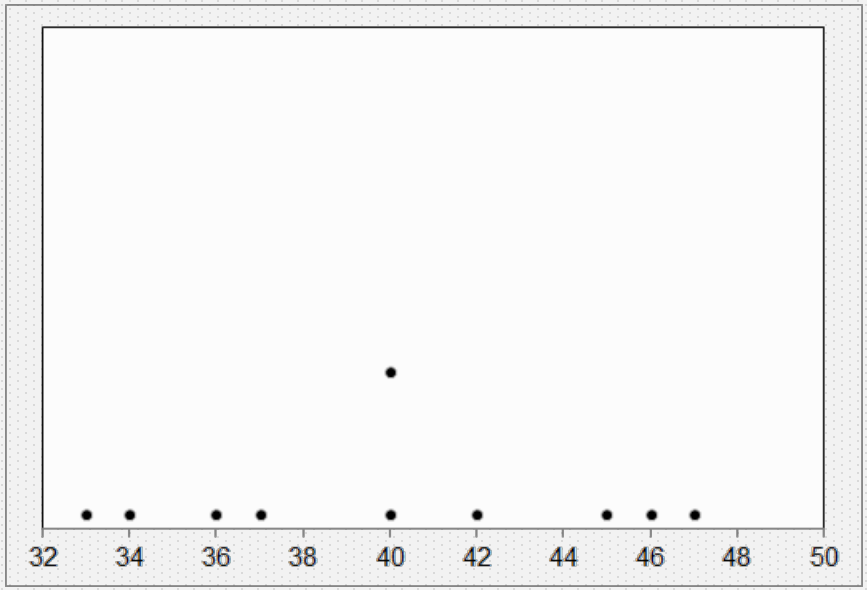
MEASURES OF VARIABILITY

Nevertheless a glance at the figure shows that they are markedly different.

In Data Set I the measurements vary only slightly from the center, while for Data Set II the measurements vary greatly



Data Set 1



Data Set 2

The Range

It is the simplest measure of variability.

The **range** of a data set is the number R defined by the formula

$$\mathbf{R} = x_{max} - x_{min}$$

where x_{max} is the largest measurement in the data set and x_{min} is the smallest.

The Range

Data Set 1:	**40**	**38**	**42**	**40**	**39**	**39**	**43**	**40**	**39**	**40**
Data Set 2:	**46**	**37**	**40**	**33**	**42**	**36**	**40**	**47**	**34**	**45**

For Data Set I the maximum is 43 and the minimum is 38, so the range is $R=43-38=5$.

For Data Set II the maximum is 47 and the minimum is 33, so the range is $R=47-33=14$.

A smaller range indicates less variability (less dispersion) among the data, whereas a larger range indicates the opposite

The Variance and the Standard Deviation

Measures of variabilities which are more elaborate and also depend on whether the data set is just a sample drawn from a much larger population or is the whole population itself.

The (Sample) variance

$$\boldsymbol{s^2} = \frac{1}{n-1}\sum_{i=1}^{n}(x-\bar{x})^2$$

The (Sample) Standard Deviation $\boldsymbol{s}$

The Variance and the Standard Deviation

Measures of variabilities which are more elaborate and also depend on whether the data set is just a sample drawn from a much larger population or is the whole population itself.

The (Population) variance

$$\boldsymbol{\sigma^2} = \frac{1}{N}\sum_{i=1}^{n}(x - \bar{x})^2$$

The (Population) Standard Deviation $\boldsymbol{\sigma}$

The Variance and the Standard Deviation

Data Set 1:	40	38	42	40	39	39	43	40	39	40
$\lvert x-\bar{x}\rvert$	0	2	2	0	1	1	3	0	1	0
$\lvert x-\bar{x}\rvert^2$	0	4	4	0	1	1	9	0	1	0
$\sum(x-\bar{x})^2$					20					

$\bar{x} = 40$

$n = 10$

$$S^2 = \frac{20}{9} = 2.\overline{2}$$

$$S = \sqrt{\frac{20}{9}} = 1.49$$

The Variance and the Standard Deviation

Data Set 2:	46	37	40	33	42	36	40	47	34	45

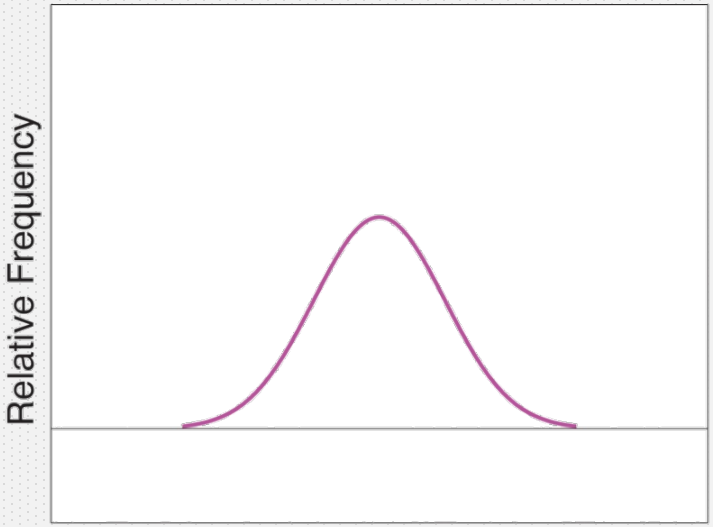
$\bar{x} = 40$

$\lvert x - \bar{x} \rvert$	6	3	0	7	2	6	0	7	6	5
$\lvert x - \bar{x} \rvert^2$	36	9	0	49	4	36	0	49	36	25
$\sum (x - \bar{x})^2$					224					

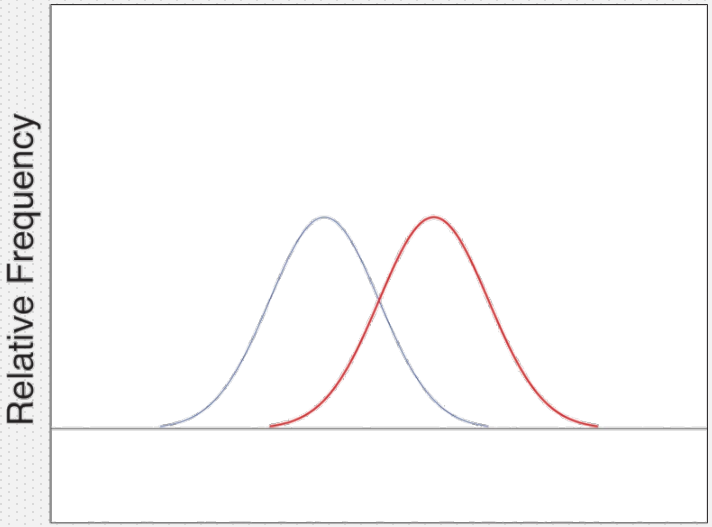
$n = 10$

$$S^2 = \frac{224}{9} = 24.\overline{8}$$

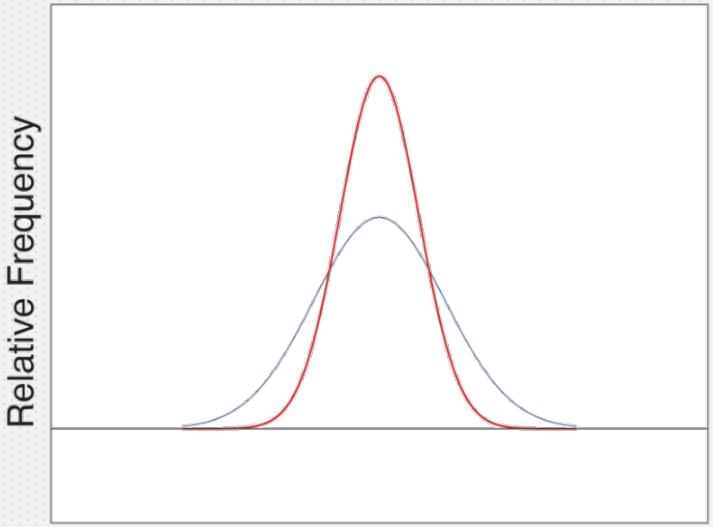
$$S = \sqrt{\frac{224}{9}} = 4.99$$



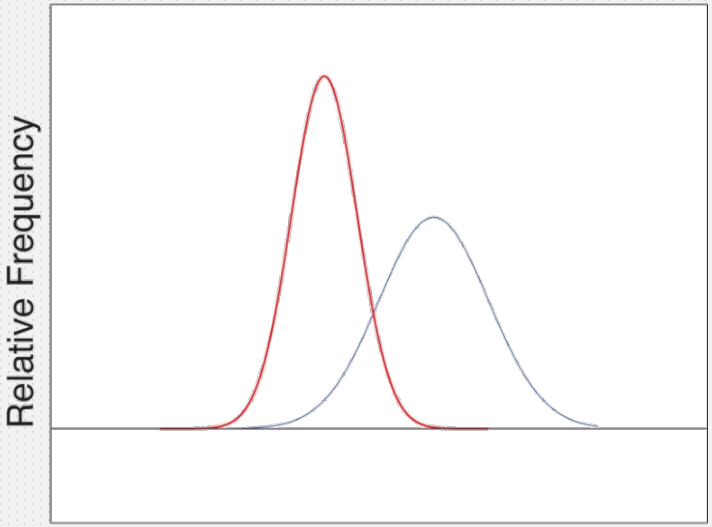
(a) Two Identical Sets



(b) Locations Differ



(c) Variabilities Differ



(d) Locations and Variabilities Differ

KEY TAKEAWAYS

- The range, the standard deviation, and the variance each give a quantitative answer to the question "How variable are the data?"

RELATIVE POSITION OF DATA

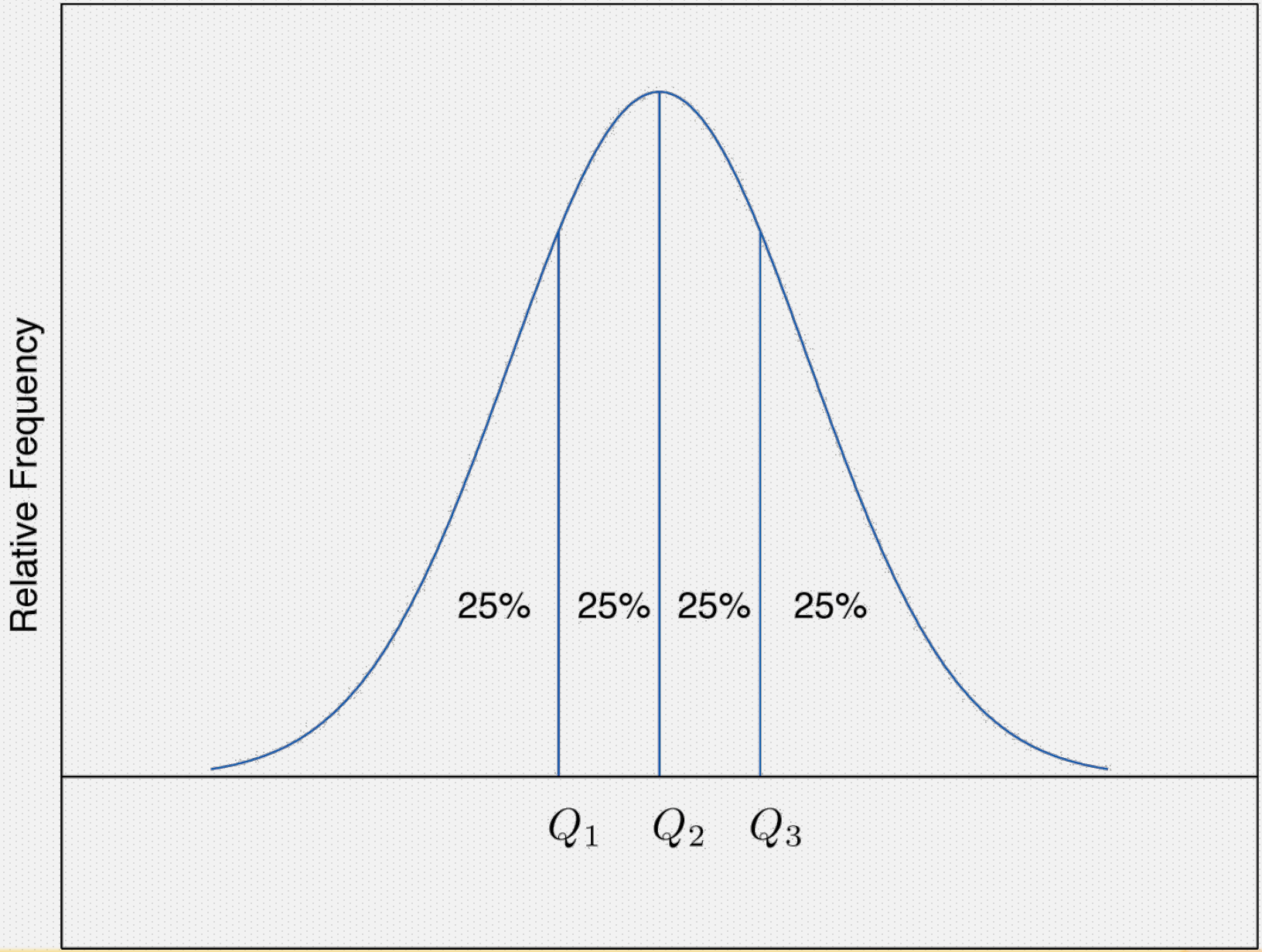
Given an observed value x in a data set, x is the **Pth percentile** of the data if the percentage of the data that are less than or equal to x is P. The number P is the **percentile rank** of x.

The **Pth percentile** cuts the data set in two so that approximately **P%** of the data lie below it and **(100−P)%** of the data lie above it

RELATIVE POSITION OF DATA

The idea of splitting **the data into halves** with **median** or **into quarters** with **quartiles** can be extended to any other fractions. **Deciles** split the data **into tenths**, **percentiles** into **hundredths**. Quantiles are a generalization of this idea to any fraction.

Data Division by Quartiles



For any data set:

1. The second quartile Q2 of the data set is its median.

2. Two subsets:

, i. the lower set: all observations that are , strictly less than Q2;

, ii. the upper set: all observations that are , strictly greater than Q2.

3. The first quartile Q1 of the data set is the median of the lower set.

4. The third quartile Q3 of the data set is the median of the upper set.

RELATIVE POSITION OF DATA

Find the quartiles of the data set of GPAs

3 1.9 2.12 3.88 1.76 2.71 3.33 2.53 1.39 3.71 4.00

Sorting:

1.39 1.76 1.9 2.12 2.53 2.71 3 3.33 3.71 3.88 4.00

Q1 Q2 Q3

RELATIVE POSITION OF DATA

How the quartiles calculated if n is even?

3 1.9 2.12 3.88 1.76 2.71 3.33 2.53 1.39 3.71

Sorting:

1.39 1.76 1.9 2.12 2.53 2.71 3 3.33 3.71 3.88

Q2

RELATIVE POSITION OF DATA

How the quartiles calculated if n is even?

1.39	**1.76**	**1.9**	**2.12**	**2.53**	**2.71**	**3**	**3.33**	**3.71**	**3.88**
1	2	3	4	**5**	6	7	8	9	10

Q2

Q2 = 50 %, n=10

0.5×(n+1) = 5.5

$$[5+0.5] \Rightarrow 2.53 + (2.71 - 2.53) \times 0.5$$

, ,

RELATIVE POSITION OF DATA

How the quartiles calculated if n is even?

1.39	**1.76**	**1.9**	**2.12**	**2.53**	**2.71**	**3**	**3.33**	**3.71**	**3.88**
1	2	3	4	5	6	7	8	9	10

10 %ile

10 %ile, n=10

0.1×(n+1) = 1.1

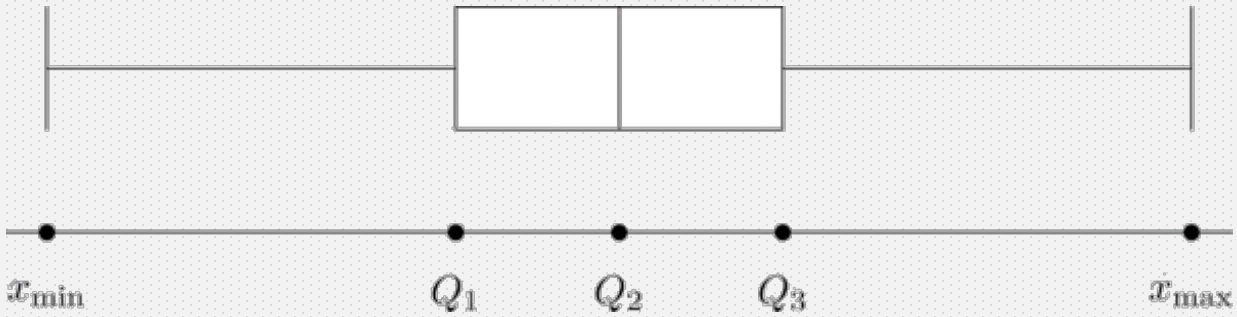
$$[1 + 0.1] \Rightarrow 1.39 + (1.76 - 1.39) \times 0.1$$

10 %ile = 1.427

Box & Whisker Plot

Five-number summary of the data set:

$$\{x_{min}, \mathrm{Q}_1, \mathrm{Q}_2, \mathrm{Q}_3, x_{max}\}$$



Note that the distance from $\boldsymbol{Q_1}$ to $\boldsymbol{Q_3}$ is the length of the interval over which the middle half of the data range. **The interquartile range ($\boldsymbol{IQR}$)** is the quantity:

$$\boldsymbol{IQR = Q_3 - Q_1}$$